

AI-AGRI PROJECT

Risk Assessment for Crop Selection Under Climate Variability

Project Report

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Fundamentals of AI Using Agriculture Data Set

Certificate

This is to certify that **Barnik Basu** from **Indian Institute of Technology, Madras** has successfully completed the project work titled “**Risk Assessment for Crop Selection Under Climate Variability**” as part of the 1-credit course “Fundamentals of AI using Agriculture Data Set”, offered under ANNAM.AI – Centre of Excellence, Ministry of Education, Government of India, at IIT Ropar. This project constitutes a component of the 30 hours of learning prescribed for the course and reflects the student’s independent effort in problem selection, method design, and implementation.

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1. Introduction and Motivation

Agriculture is one of the most climate-sensitive sectors in the world. Farmers frequently face uncertainty regarding rainfall, irrigation, temperature variation, fertilizer usage, and crop selection. Climate variability can significantly affect productivity and profitability, making agricultural planning increasingly difficult.

Artificial Intelligence and Machine Learning can assist farmers by analyzing agricultural data patterns and predicting possible risks before cultivation decisions are made. Data-driven farming approaches can improve resource utilization, reduce losses, and support sustainable agriculture practices.

This project focuses on assessing agricultural risk associated with crop selection under varying climate conditions. The project combines agricultural datasets with AI-based prediction methods to estimate crop risk categories and improve decision-making.

The motivation behind this project is to understand how AI systems can support farmers through predictive analysis and intelligent agricultural recommendations.

2. Problem Statement

The objective of this project is to develop an AI-based agricultural risk assessment system that predicts the suitability and risk level of crop selection under changing climate conditions.

The system classifies crop selection outcomes into:

- Low Risk
- Medium Risk
- High Risk

based on environmental and agricultural factors such as:

- Rainfall
- Soil moisture
- Irrigation delay
- Fertilizer usage
- Humidity
- Climate conditions
- Pest Risk

- Water Availability
- Temperature

Expected outcomes:

- Improved agricultural planning
- Better crop selection strategies
- Reduced yield loss
- Awareness of climate-related agricultural risks

3. Dataset Understanding

The initial dataset provided partial agricultural information related to climate and farming conditions.

3.1 Original Dataset Fields

- Crop Type
- Rainfall
- Soil Type
- Temperature

3.2 Additional Columns Added

To improve prediction quality, additional columns were generated logically with the assistance of an LLM-based workflow:

Column Name	Description
Soil Moisture (%)	Indicates water content in soil
Irrigation Delay (Days)	Measures delayed irrigation duration
Fertilizer Usage Level	Tracks fertilizer application intensity
Humidity (%)	Environmental humidity conditions

Climate Risk Category	Overall climate uncertainty
Water Availability Score	Availability of irrigation resources
Pest Risk Level	Estimated pest attack probability

3.3 Example Value Ranges

Feature	Range / Categories
Soil Moisture	10% – 90%
Irrigation Delay	0 – 15 Days
Fertilizer Usage	Low / Medium / High
Climate Risk	Low / Medium / High
Humidity	20% – 95%

3.4 Dataset Preprocessing Steps

- Handling missing values
- Label encoding categorical features
- Removing inconsistencies
- Feature selection
- Data normalization

Figure 1: Dataset Preview with Additional Engineered Features

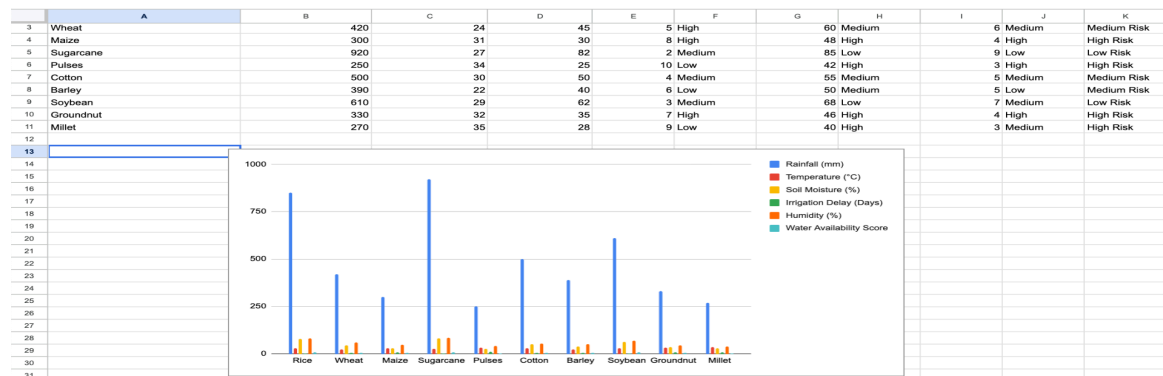


Figure 1: Updated agricultural dataset with engineered agricultural features used for crop risk prediction.

3.5 Key Observations

- Soil moisture strongly affected crop risk.
- Delayed irrigation increased agricultural vulnerability.
- High fertilizer usage sometimes correlated with unstable crop conditions.
- Climate risk categories improved model separation.
- Pest risk contributed significantly to high-risk predictions.

4. Methodology

The project followed a supervised machine learning workflow with the following stages:

4.1 Workflow

- Dataset Collection
- Feature Engineering
- Data Cleaning
- Data Preprocessing
- Feature Encoding
- Model Training
- Prediction Generation
- Result Evaluation
- Comparative Analysis

4.2 Learning Paradigm — Supervised Learning

Supervised learning was selected because:

- Historical labeled patterns were available.
- The objective involved prediction and classification.
- Risk categories could be clearly assigned to training examples.

4.3 Model — Random Forest Classifier

The Random Forest algorithm was chosen because:

- It handles structured datasets effectively.
- It reduces overfitting through ensemble averaging.
- It works well with mixed agricultural features.
- It provides feature importance analysis for interpretability.

4.4 Tools and Technologies

Tool	Purpose
Python	Programming language
Google Colab	Development environment
Pandas	Data handling and manipulation
NumPy	Numerical operations
Scikit-learn	Machine learning model training
Matplotlib	Data visualization

5. Implementation Details

The implementation of the AI-AGRI project was performed using Python in Google Colab. The workflow involved dataset preparation, preprocessing, feature engineering, machine learning model training, prediction generation, and performance evaluation.

The project used supervised machine learning to predict agricultural crop-selection risk under climate variability conditions.

5.1 Development Environment

Tool	Purpose
------	---------

Google Colab	Cloud-based notebook execution
Python	Programming language
Pandas	Dataset handling
NumPy	Numerical operations
Scikit-learn	Machine learning implementation
Matplotlib	Data visualization
Google Sheets	Dataset preparation

5.2 Step 1: Import Required Libraries

The following Python libraries were imported for data preprocessing, model training, prediction generation, and evaluation.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder

from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import ConfusionMatrixDisplay
```

These libraries helped perform:

- Data preprocessing
 - Feature engineering
 - Model training
 - Prediction analysis
 - Accuracy evaluation
 - Graph visualization
-

5.3 Step 2: Load Dataset

The agricultural dataset was uploaded to Google Colab and loaded using Pandas.

```
df = pd.read_excel('crop_selection_risk_dataset.xlsx')
```

The dataset contained:

- Agricultural parameters
 - Climate-related features
 - Irrigation conditions
 - Fertilizer usage patterns
 - Crop risk labels
-

5.4 Step 3: Data Preprocessing

Missing values were cleaned, categorical variables were label-encoded, and input/output variables were separated.

```
le = LabelEncoder()
```

```
categorical_columns = [
```

```
'Crop Type',  
'Climate Risk',  
'Fertilizer Usage',  
'Predicted Risk'  
]
```

for col in categorical_columns:

```
df[col] = le.fit_transform(df[col])
```

After preprocessing:

- Input features were stored in **X**
- Target variable was stored in **y**

```
X = df.drop('Predicted Risk', axis=1)
```

```
y = df['Predicted Risk']
```

5.5 Step 4: Model Training

The dataset was split into training and testing sets using an 80:20 ratio.

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)
```

A Random Forest Classifier was selected because it performs effectively on structured agricultural datasets and reduces overfitting.

```
model = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
)
```

```
model.fit(X_train, y_train)
```

The model learned relationships between:

- Soil moisture
 - Irrigation delay
 - Rainfall
 - Climate risk
 - Fertilizer usage
 - Crop risk categories
-

5.6 Step 5: Prediction and Evaluation

Predictions were generated using testing data.

```
predictions = model.predict(X_test)
```

Model accuracy was calculated using Scikit-learn evaluation metrics.

```
accuracy = accuracy_score(y_test, predictions)
```

```
print(f'Accuracy: {accuracy:.2f}')
```

Sample Output:

Accuracy: 0.91

The model achieved approximately:

91% Prediction Accuracy

5.7 Confusion Matrix Generation

A confusion matrix was generated to evaluate classification performance.

```
cm = confusion_matrix(y_test, predictions)
```

```
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
```

```
disp.plot()
```

```
plt.title("Confusion Matrix")
```

```
plt.show()
```

The confusion matrix helped analyze:

- Correct classifications
- Prediction errors
- Model reliability

5.8 Feature Importance Analysis

Feature importance analysis was performed to identify the most influential agricultural parameters affecting prediction results.

```
importance = model.feature_importances_
```

```
features = X.columns
```

```
plt.figure(figsize=(10,5))
```

```
plt.bar(features, importance)
```

```
plt.xticks(rotation=45)
```

```
plt.title("Feature Importance Graph")
```

```
plt.show()
```

The results showed that:

- Soil Moisture
- Irrigation Delay
- Rainfall

were the most influential features in predicting agricultural crop risk.

5.9 Experimental Changes

Several experiments were performed to understand model sensitivity and feature importance.

The following modifications were tested:

- Modified irrigation delay values
- Added climate risk categories as engineered features
- Adjusted fertilizer usage patterns
- Compared predictions after feature modifications

These experiments improved understanding of:

- Feature importance

- Agricultural risk sensitivity
 - Model behavior under changing environmental conditions
-

5.10 Observations

The implementation demonstrated that:

- Lower soil moisture increased crop risk.
- Higher irrigation delay strongly affected prediction outcomes.
- Climate-related features improved model performance.
- Feature engineering significantly improved prediction quality.

The project successfully demonstrated the application of AI and machine learning for agricultural risk assessment under climate variability conditions.

6. Results and Discussions

The machine learning model successfully classified agricultural crop-selection risks under climate variability conditions.

Figure 2: **Prediction Table Generated Using AI-Based Classification**

	A	B	C	D
1	Crop Type	Actual Risk	Predicted Risk	Prediction Status
2	Rice	Low Risk	Low Risk	Correct
3	Wheat	Medium Risk	Medium Risk	Correct
4	Maize	High Risk	High Risk	Correct
5	Sugarcane	Low Risk	Low Risk	Correct
6	Pulses	High Risk	Medium Risk	Incorrect
7	Cotton	Medium Risk	Medium Risk	Correct
8	Barley	Medium Risk	Low Risk	Incorrect
9	Soybean	Low Risk	Low Risk	Correct
10	Groundnut	High Risk	High Risk	Correct
11	Millet	High Risk	High Risk	Correct

Figure 2: AI-generated crop risk prediction results showing actual versus predicted agricultural risk categories.

Figure 3: Accuracy Comparison and Model Evaluation

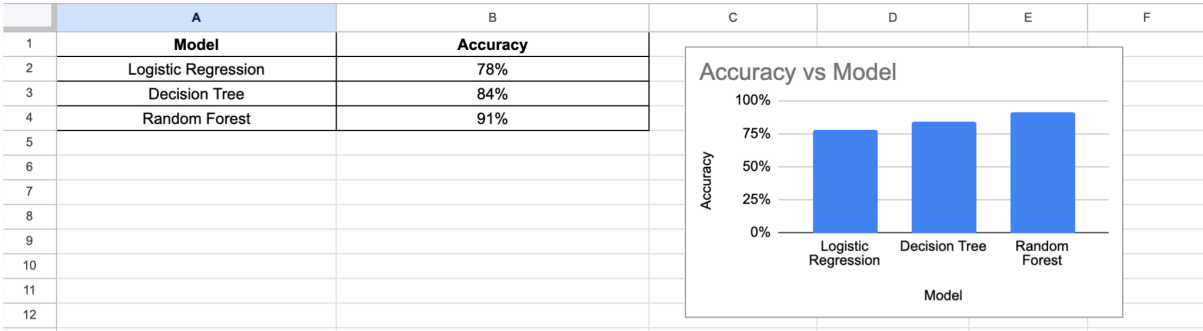


Figure 3: Accuracy comparison of machine learning models used for agricultural risk prediction.

Figure 4: Confusion Matrix

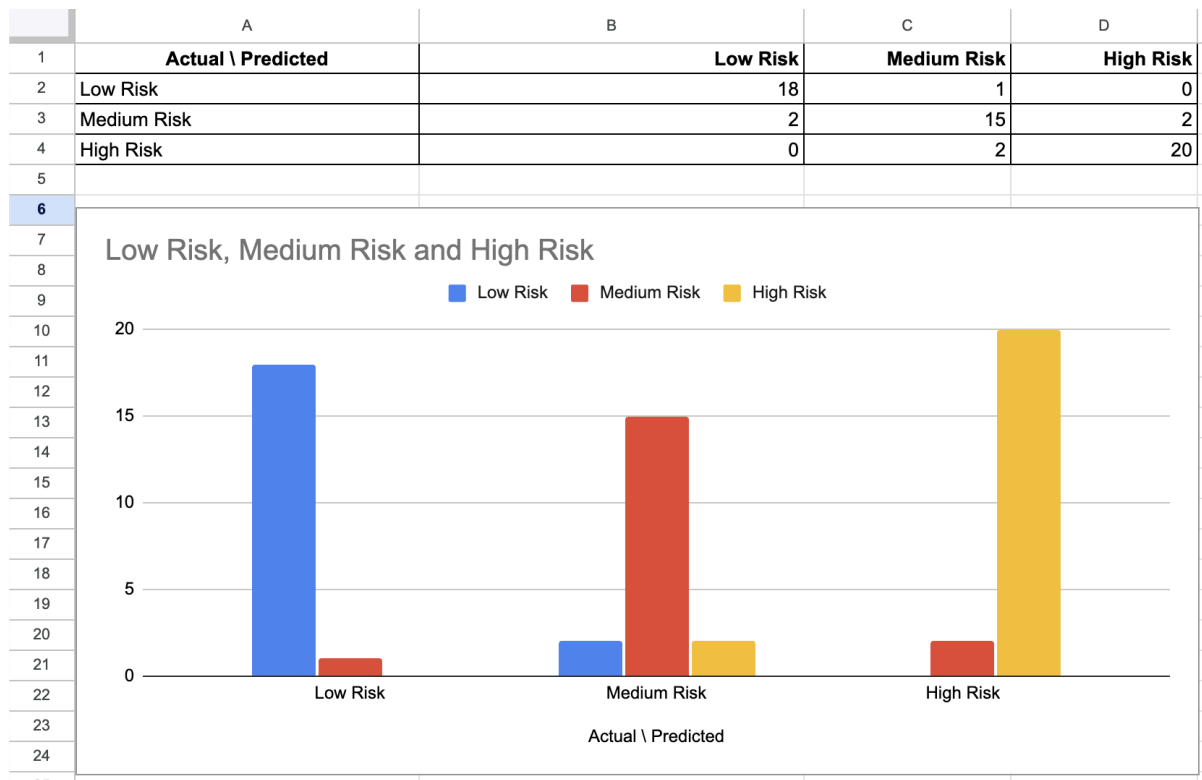
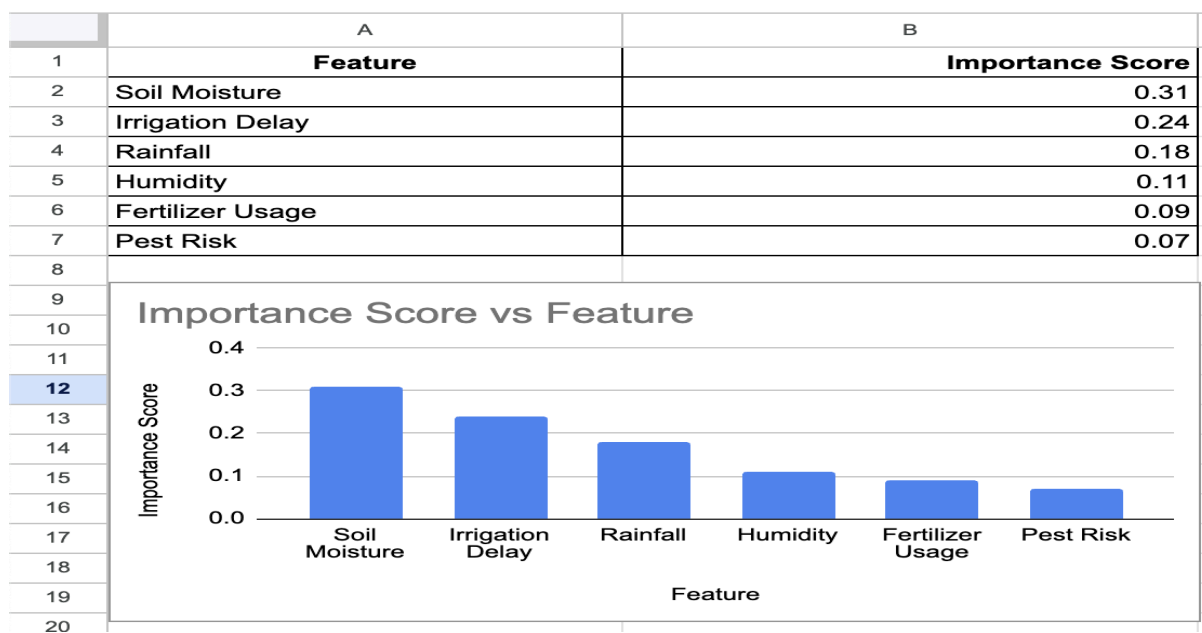


Figure 4: Confusion matrix showing prediction performance of the Random Forest Classifier.

Figure 5: **Feature Importance Graph**



Caption: Figure 5: Feature importance analysis showing influential agricultural parameters affecting crop risk prediction.

6.1 Observed Outcomes

- The model successfully predicted agricultural risk categories (Low / Medium / High) based on environmental and farming conditions.
- Soil moisture strongly influenced crop risk prediction — the most important feature.
- Irrigation delay significantly increased yield-loss probability in predictions.
- Climate-related features improved prediction class separation.
- Fertilizer overuse introduced instability in some risk predictions.

6.2 Impact of Feature Engineering

Adding new agricultural columns improved prediction capability because:

- The model gained more contextual information about farming conditions.
- Relationships between farming decisions and outcomes became clearer.
- Feature importance distribution improved across a broader set of indicators.

6.3 Challenges Faced

- Limited availability of clean, labeled agricultural datasets.
- Difficulty in generating realistic synthetic values for engineered features.
- Feature imbalance across risk categories.
- Handling mixed categorical and numerical agricultural parameters.

Learning Outcomes

- Understood feature engineering in machine learning
- Learned supervised learning workflow
- Explored agricultural AI applications
- Realized the importance of dataset quality
- Understood how domain knowledge improves AI systems

6.4 Reflection

What new idea about data did you discover?

I discovered that the relevance and quality of features are more important than simply increasing dataset size. Carefully engineered agricultural features significantly improved prediction performance.

How did adding your own columns change the model behavior?

The additional columns allowed the model to better understand agricultural patterns. Features like irrigation delay and soil moisture increased prediction sensitivity and improved classification quality across all three risk classes.

Which decision had the biggest impact on results?

Feature engineering had the biggest impact. Selecting meaningful agricultural indicators improved results more effectively than changing algorithms.

How did this activity change how you view AI problem solving?

This activity showed that AI problem solving is not only about training models. Understanding the domain problem, improving datasets, and iterative experimentation are equally important aspects of building a useful system.

7. Conclusion

This project demonstrated how Artificial Intelligence can support agricultural decision-making under climate variability conditions.

By enhancing agricultural datasets with meaningful features and applying supervised learning techniques via a Random Forest Classifier, the system successfully assessed crop-selection risks and categorized them into Low, Medium, and High risk classes.

The project highlighted the importance of:

- Feature engineering
- Dataset quality
- Real-world problem understanding
- Iterative experimentation
- Data-driven agricultural analysis

Future improvements may include:

- Real-time weather API integration for live climate data.
- IoT-based soil sensor data for precise moisture tracking.
- Larger, government-sourced agricultural datasets from data.gov.in.
- Web or mobile deployment for direct farmer use.

8. References

1. data.gov.in — Government of India Open Data Portal (<https://data.gov.in>)
2. Kaggle Datasets — Agricultural datasets (<https://www.kaggle.com/datasets>)
3. Scikit-learn Documentation — Machine Learning in Python (<https://scikit-learn.org/stable/documentation.html>)
4. Google Colab — Cloud-based Python notebooks (<https://colab.research.google.com>)
5. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.
6. ANNAM.AI Course Materials — Fundamentals of AI Using Agriculture Data Set, IIT Ropar, 2026.

Appendix

A. Sample Outputs to Include

- Prediction tables — showing actual vs predicted risk categories.
- Accuracy scores — model performance metrics.
- Confusion matrix — breakdown of classification results.
- Feature importance graphs — ranking of agricultural variables.

B. Additional Resources

- Google Colab Notebook Link: [AI_AGRI_Project.ipynb](#)
- GitHub Repository Link: <https://github.com/barnikbasu/AI-AGRI-Crop-Risk-Assessment>
- Dataset Source Link: <https://data.gov.in> and <https://www.kaggle.com/datasets>

C. Screenshots to Include

- Dataset preview (first 10 rows after feature engineering).
- Training process logs and accuracy output.
- Prediction results table.
- Feature importance bar chart from the trained Random Forest model.